Online

Shopping Database

Student ID	Student Name
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Customer Table

Create Table:

```
Create table Customer (CID char(4) Priamry key, CName varchar(20) Not Null, CPhone int null, CCity varchar(20) Not Null, CPostal_Code int not null, OrderID char(5) unique, CAge int); --Customer table
```

Insert Values Into Table:

- ```
  INSERT INTO Customer Values ('C100', 'Ahmad', 90990889,'Oman', 403, 'O122' ,20); --
  ```

- ```
  INSERT INTO Customer Values ('C200',  'Reem',  90876050, 'USD' ,500,  'Q334', 13);
  ```

- ```
  INSERT INTO Customer Values ('A101',  'Ali, 98765432, 'UK', 121,  'O890', 44);
  ```

- ```
  INSERT INTO Customer Values ('A201',  'Mlak',  98789808, 'India',421,  'Q546', 30);
  ```

CID	CNAME	CPHONE	CCITY	CPOSTAL_CODE	ORDERID	CAGE
C100	Ahmad	90990889	Oman	403	O122	20
C200	Reem	90876050	USD	500	Q334	13
A101	Ali	98765432	UK	121	O890	44
A201	Mlak	98789808	India	421	Q546	30

4 rows returned in 0.01 seconds Download

ORDER Table

Create Table:

```
Create table CORDER (OrderID char(5) Primary key, ODetailsID char(5)
unique, OpaymentDate Date Not Null, OState varchar(20) Not Null,
PamentMethods varchar(20) Not null ,OQuantityint int, OType
varchar(50));
```

Insert Values Into Table:

- ➔ `INSERT INTO CORDER Values ('O122','D100', '29/sep/2022', 'Charged', 2,'Accessories', 'PayPal');`
- ➔ `INSERT INTO CORDER Values ('Q334','D200', '5/June/2023', 'Shipped', 4,'Phone Protector','Visa' );`
- ➔ `INSERT INTO CORDER Values ('O890','D500', '9/July/2022', 'Received', 3, 'Phone Protector', 'Cash');`
- ➔ `INSERT INTO CORDER Values ('Q546','D110', '7/Aug/2020', 'Received', 1, 'Accessories', 'Credit');`

ORDERID	ODETAILSID	OPAYMENTDATE	OSTATE	OQUANTITYINT	OTYPE	PAMENTMETHODS
O122	D100	29-Sep-2022	Charged	2	Accessories	PayPal
Q334	D200	05-Jun-2023	Shipped'	4	Phone Protector	Visa'
O890	D500	09-Jul-2022	Received	3	Phone Protector	Cash
Q546	D110	07-Aug-2020	Received	1	Accessories	Credit

4 rows returned in 0.00 seconds Download

5. Preform any two(2) Nested functions using any of above created tables.	0.5

1- `Select concat(CID, substr(CName,1,1) ) newID from customer;`

NEWID
C100A
C200R
A101A
A201M

4 rows returned in 0.00 seconds Download

2- `Select substr(CName, 1 , instr('Welcome customer', 'W') Customer_Name from customer where CCity IN ('Oman', 'India') ;`

CUSTOMER_NAME
A
M

2 rows returned in 0.01 seconds Download

6. Create any correlated Subqueries	0.5

```
1- SELECT CName,Cphone,CCity FROM Customer WHERE OrderID= (SELECT
   OrderID FROM Customer WHERE CName = 'Reem');
```

CNAME	CPHONE	CCITY
Reem	90876050	USD

1 rows returned in 0.01 seconds Download

```
2- SELECT PName, Price FROM PRODUCTS WHERE Price < (SELECT
   AVG(Price)FROM PRODUCTS);
```

7. Create any complex view	0.5

```
CREATE OR REPLACE VIEW customer_view AS SELECT * FROM
orders WHERE customer_id IN (SELECT customer_id FROM
customers WHERE city = 'USD');
```

Part II: (Self Study) Using NOSQL(MongoDB), write the code on paper to perform the below tasks:

1. Create one of the above tables	1

```
const mongoose = require('mongoose');

const customerSchema = new mongoose.Schema({

  CID: String,

  CName: String,

  CPhone: int,

  CCity:string,

  CPostal_Code: int,

  OrderID: String,

  CAge: int,

});

const Customer = mongoose.model('Customer', customerSchema);

module.exports = Customer;
```

2. Insert at least 3 records in the created table	0.75

```
db.customer.insertMany([

{

  "CID": "C100", CName: "Ahmad", CPhone: 90990889, CCity: "Oman", CPostal_Code: 403, OrderID:
"O122", CAge: 20,

}

{

  "CID": "C200", CName: "Reem", CPhone: 90876050, CCity: "USD", CPostal_Code: 500, OrderID:
"Q334", CAge: 13

}
```

```
{

 "CID": "A101", CName: "Ali", CPhone: 98765432, CCity: "UK", CPostal_Code: 121, OrderID: "O890",
CAge: 44

}

{

  "CID": "A201", CName: "Malak", CPhone: 98789808,

  CCity: "india", CPostal_Code: 421, OrderID: "Q546", CAge: 30

}
```

3. Display columns with two conditions	0.75

```
const customer = db.customer.find({

  $and: [

    {age: {$gt: 30}},

    {city: "New York"}

  ]

});
```

4. Update a record with a condition	0.75

```
db.customer.updateOne(

  { CID:"A101" },

  {

    $set:

      {

  "CID": "A101",
```

```
  CName: "Hamed",
  CPhone: 98765432,
  CCity: "Canada",
  CPostal_Code: 190,
  OrderID: "O110",
  CAge: 66
      }
  },
  { upsert: true }
)
```

1. Delete a record with a condition	0.75

```
db.customer.deleteOne({ CID: "A101" })
```

